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B. E. (Fourth Semester) Examination, Nov.-Dec. 2016

(New Scheme)

(Elect. Engg. Branch)

ELECTROMAGNETIC THEORY

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory. Solve any two parts from (b), (c) and (d).*

1. (a) State the Coulomb's law. 2

(b) Find $\overline{E}$ at P (1, 5, 2) in free space if a point charge

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of $6\mu c$ is located at Q (0, 0, 1), a uniform line charge of 180 nc/m lies along the x-axis and a uniform sheet of charge equal to 25 nc/m^2 lies in the plane $x = -1$. 7

(c) A charge $Q_1 = 8\mu c$ is located at $P_1(2, 5, 8)$ while $Q_2 = -5\mu c$ is at $P_2(6, 15, 8)$ in free space. 7

(i) Find the force on Q_2 due to Q_1.

(ii) Find the co-ordinates of P_3 if a charge Q_3 experiences a total force $\overline{F}_3 = 0$ at P_3.

(d) Determine the expression for the electric field $\overline{E}$ at a general point P due to charge uniformly distributed along an infinite straight line with charge density. 7

2. (a) State the divergence theorem. 2

(b) A charge distribution with spherical symmetry has density,

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$$\rho_v \left(\frac{\rho_0 r}{a}\right) \quad 0 \le r \le a$$

$$= 0 \quad r > a$$

Determine $\bar{E}$ everywhere. 7

(c) Explain Gauss law with its application. 7

(d) A circular disc, $\rho \le 1$ cm, $z = 0$, carries a sureface charge density $\rho_s = \frac{1}{r}$ nc/m²

find V at $P(0, 0, z)$ if $V = 0$ at infinity.

3. (a) Write the following equation : 2

(i) Continuity euqation for current

(ii) Poisson's equation

(b) Explain boundary conditions for perfect dielectric materials. 7

(c) A point charge of 25 nC is located at in free space

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at $P(2, -3, 5)$ and a perfectly conducting plane is at $z = 2$. 7

Find :

(i) V at (3, 2, 4)

(ii) $\bar{E}$ at (3, 2, 4)

(d) Conducting cylinders at $\rho = 1.6$ cm and $\rho = 5$ cm in free space are held at potentials of 80 V and – 40 V, respectively : 7

(i) Find V and $\bar{E}$ at $\rho = 2$ cm.

(ii) Find the surface at which $V = 0$

(iii) Find the surface charge density on each cylinder.

4. (a) What is magnetic susceptibility? 2

(b) A charge particle of mass 2 kg and charge 3 C starts at point (1, –2, 0) with velocity $(4\hat{a}_x + 2\hat{a}_z)$ m/s in an electric field $12\hat{a}_x + 10\hat{a}_y$ V/M. At time $t = 1$ sec.

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Determine :

(i) The acceleration of the particle

(ii) Its velocity

(iii) Its kinetic energy

(c) State and prove Stoke's theorem. 7

(d) A curent element $I\,\overline{dL} = 10^{-3}\left(2\hat{a}_x + 4\hat{a}_y - \hat{a}_z\right)$

A.M. located at A (–5, 3, 2) produces a field $\overline{dH}$ at B (3, –4, 3).

(i) Give a unit vector in the direction of $\overline{dH}$ at B.

(ii) Find $\left|\overline{dH}\right|$ at B.

(iii) What direction should $I\,\overline{dL}$ have at A so that $\overline{dH} = 0$ at B. 7

5. (a) What is Poynting vector? 2

(b) Explain the following : (any two) 7

(i) Faraday's law of electromagnetic induction.

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(ii) Displacement current and displacement current density

(iii) Statically and dynamically induced emf.

(c) The sides of a square loop in the $z = 0$ plane are located at $x = \pm 0{\cdot}6$ m and $y = \pm 0{\cdot}6$ m. There exists a uniform time varying magnetic field given by

$$\overline{B} = \left(0{\cdot}2\hat{a}_x - 0{\cdot}4\hat{a}_y + 0{\cdot}8\hat{a}_z\right)\cos 200t \text{ wb/m}^2$$

If the total resistance of the square loop is $1\,\text{k}\Omega$. Find the current through the loop. 7

(d) Assume a homogeneous material of infinite extent with $\varepsilon = 2\times10^{-1}$ F/M,

$\mu = 1{\cdot}25\times10^{-5}$ H/M, $\sigma = 0$ let

$$\overline{E} = 400\cos\left(10^9 t - kz\right)\hat{a}_y \text{ V/M}$$

use Maxwell's equation to find

$\overline{D}, \overline{B}, \overline{H}$ and K. 7

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